L4 Safety Island — the six declared behaviours

The island's node graph (phase 5 of the series), lit once per behaviour of §3.1

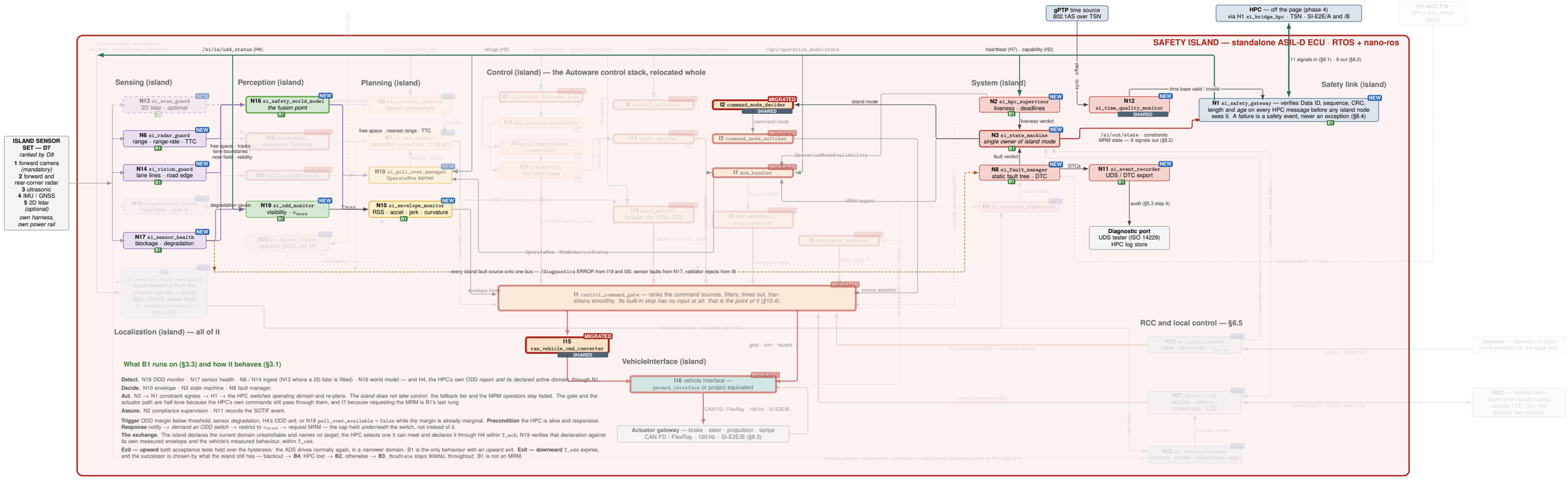
B1	Degrade the ADS when the ODD cannot be held	The island constrains and demands an ODD switch; the HPC executes. The only behaviour with an upward exit
B2	Keep the vehicle in its lane after loss of the HPC	The island takes authority and holds the lane for a bounded window while the lane is still observable
B3	Pull over if the ODD continues to fail	A placed stop in a verified refuge, nominated by the HPC or found by the island
B4	Emergency stop when the island is blind	An unplaced stop with no exteroceptive sensor at all: the island brakes and declines to steer
B5	Tele-operation from the remote command centre	A remote driver commands over the island's own link, on the sensing it streams; the island's guards stay armed. *From standstill, or taken over from B1–B3 (§6.6)*
B6	Manual joystick control without streamed sensing	An operator in sight drives at a creep on the local control port; no sensor is required. *Stage 1, from standstill (§6.6)*

On every page the drawing is the same. What the behaviour runs on — the nodes named in its §3.3 row — and the SHARED nodes every behaviour runs through are in **full colour**; what it only passes through by §3.1 is **half-tone**; the rest of the island is **faded**. An edge is drawn at the weaker of its two ends, and every lit node keeps at least one lit edge in and one out. The lower-left panel of each page lists the behaviour's Detect · Decide · Act · Assure nodes and its trigger, precondition, response and exit.

Derived from `L4SafetyIsland_SoftwareDesign.md`, version 20261023, without modifying it.

Phase 5 · B1 — Degrade the ADS when the ODD cannot be held

When the operating envelope is no longer satisfied, require the ADS to move to an operating domain it *can* satisfy — progressively constraining it, and demanding that it constrain itself, until it does — before any authority is taken.
The phase-5 drawing, unchanged: what B1 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1

SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B1, degrade the ADS

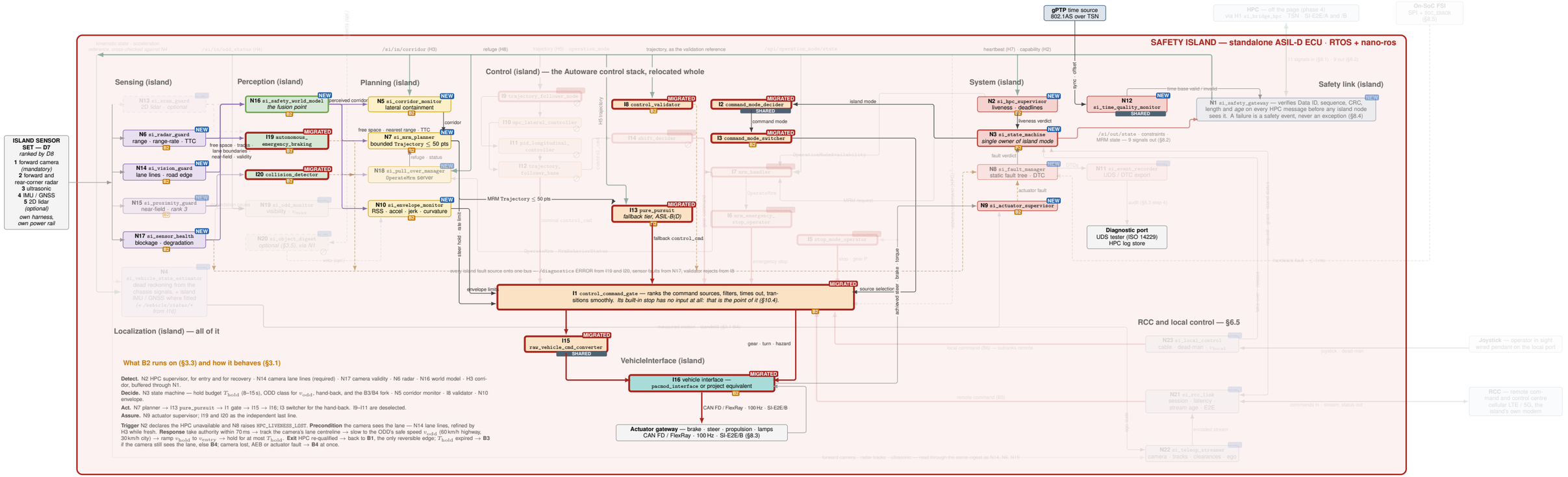
full colour — named for B1 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B1's path by §3.1, but not named in §3.3

faded — not used by B1. An edge is drawn at the weaker of its two ends. Only B1's own tag is drawn on each box.

Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded 8–15 s window, first down to the ODD's safe speed — while the camera still sees the lane.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of `ArchitectureMain.pdf`

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Where each node came from

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B2 · SHARED · B2 · ⊘

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B2, lane-keep hold

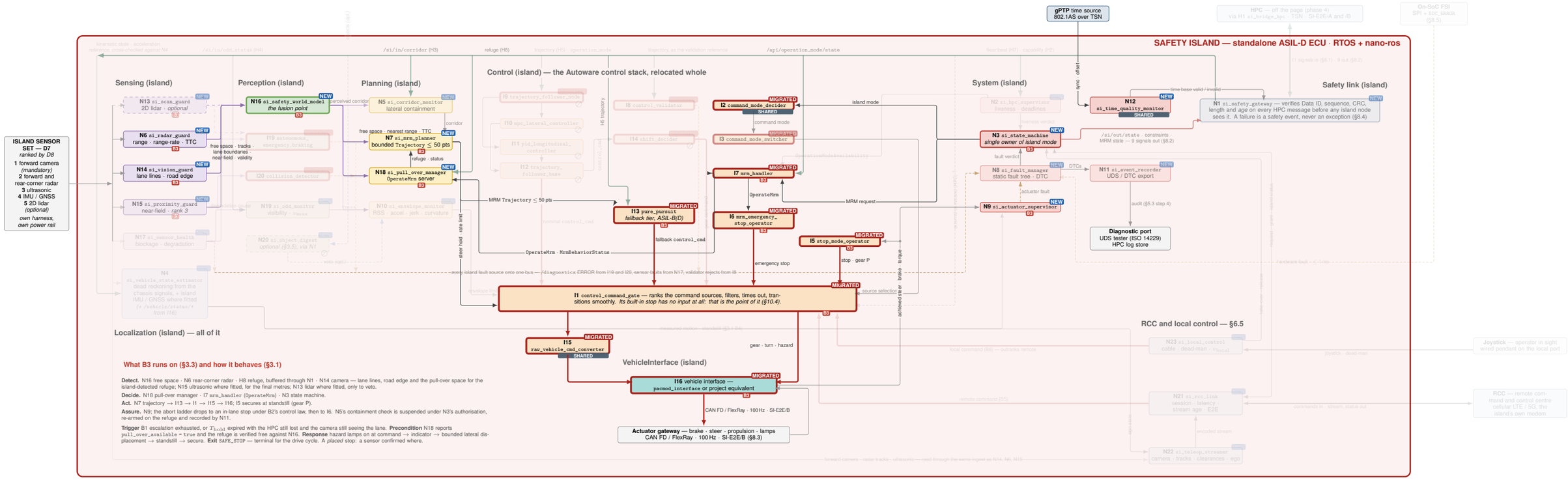
full colour — named for B2 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B2's path by §3.1, but not named in §3.3

faded — not used by B2. An edge is drawn at the weaker of its two ends. Only B2's own tag is drawn on each box.

Phase 5 · B3 — Pull over if the ODD continues to fail

When degradation does not restore the envelope within a declared deadline, bring the vehicle out of the running lane to a standstill in a refuge — nominated by the HPC, or found by the island's camera.
The phase-5 drawing, unchanged: what B3 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1 · B2 · B3 · B4 · B5

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

HPC-sourced, verified by N1
island sensor data
control command into the gate
fault · `/diagnostics`
actuator feedback
state · mode · decision
optional (§3.5)

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In this view — B3, pull over

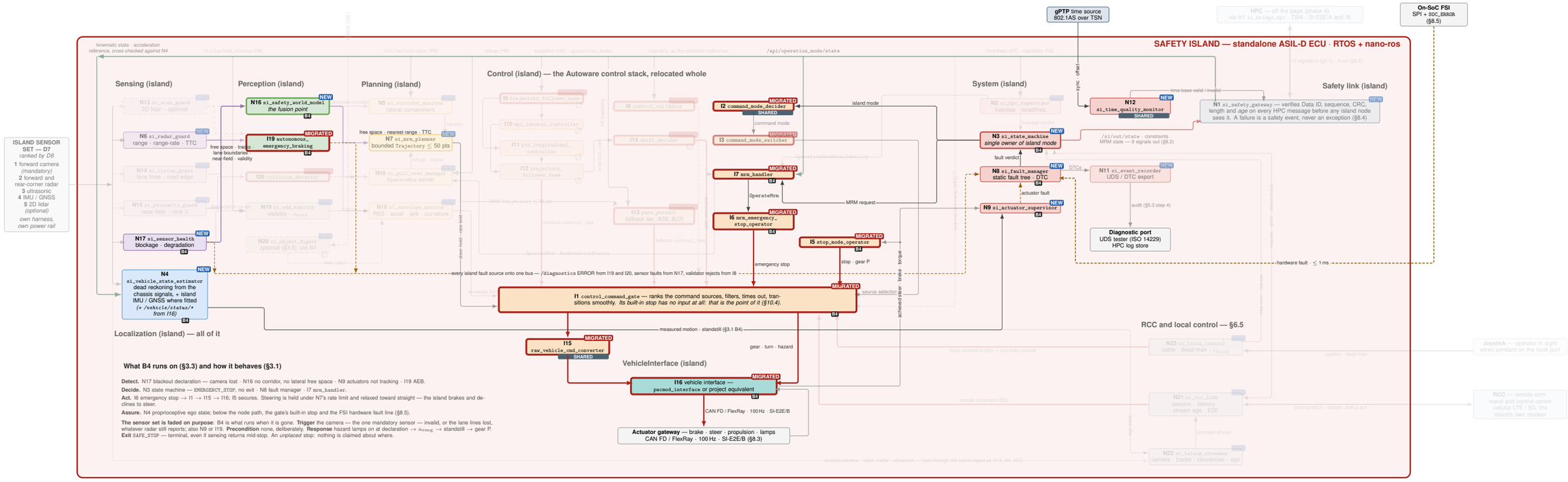
full colour — named for B3 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B3's path by §3.1, but not named in §3.3

faded — not used by B3. An edge is drawn at the weaker of its two ends. Only B3's own tag is drawn on each box.

Phase 5 · B4 — Emergency stop when the island is blind

When the island has lost the sensing that B2 and B3 both depend on — the camera — decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre.
The phase-5 drawing, unchanged: what B4 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of ArchitectureMain.pdf

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Where each node came from

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Behaviours each node enables (§3.3)

one fixed slot per behaviour B4
SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · /diagnostics
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B4, blind emergency stop

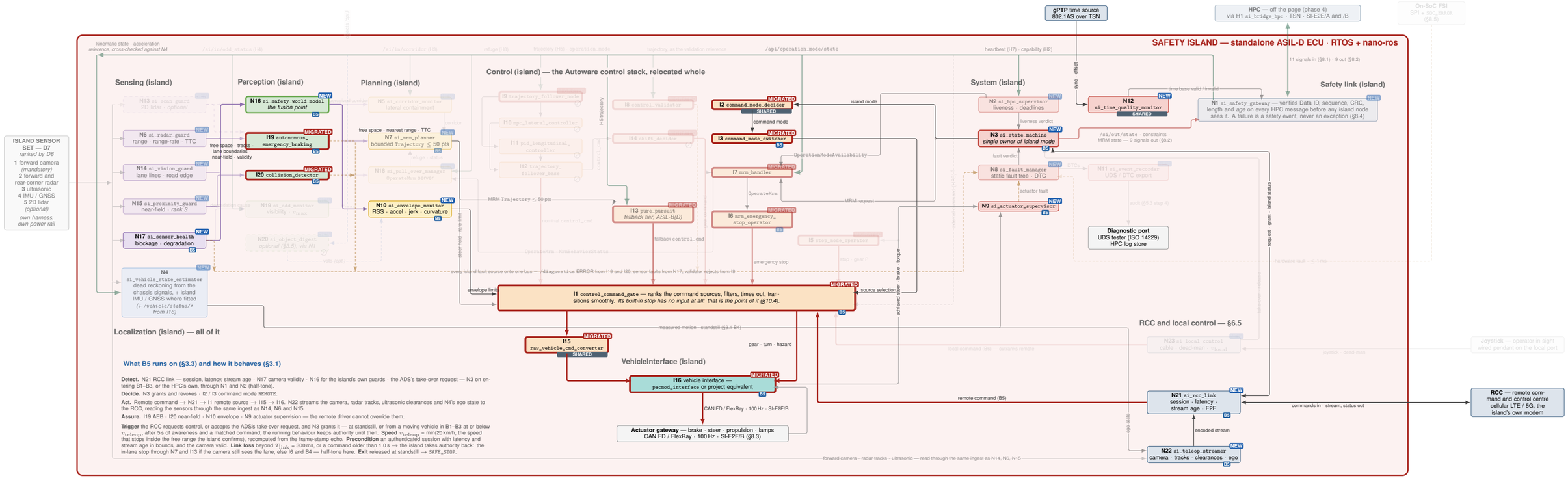
full colour — named for B4 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B4's path by §3.1, but not named in §3.3

faded — not used by B4. An edge is drawn at the weaker of its two ends. Only B4's own tag is drawn on each box.

Phase 5 · B5 — Tele-operation from the remote command centre

A remote driver at the RCC drives the vehicle over the island's own communication module, on the camera, radar and ultrasonic the island streams — while the island keeps its guards armed and stops the vehicle itself if the link fails.
The phase-5 drawing, unchanged: what B5 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Detailed design: §6.6 — from standstill, or taken over from a moving vehicle in B1–B3 on the ADS's request.*



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Where each node came from

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B5 · SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

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Messages

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- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B5, tele-operation from the remote command centre

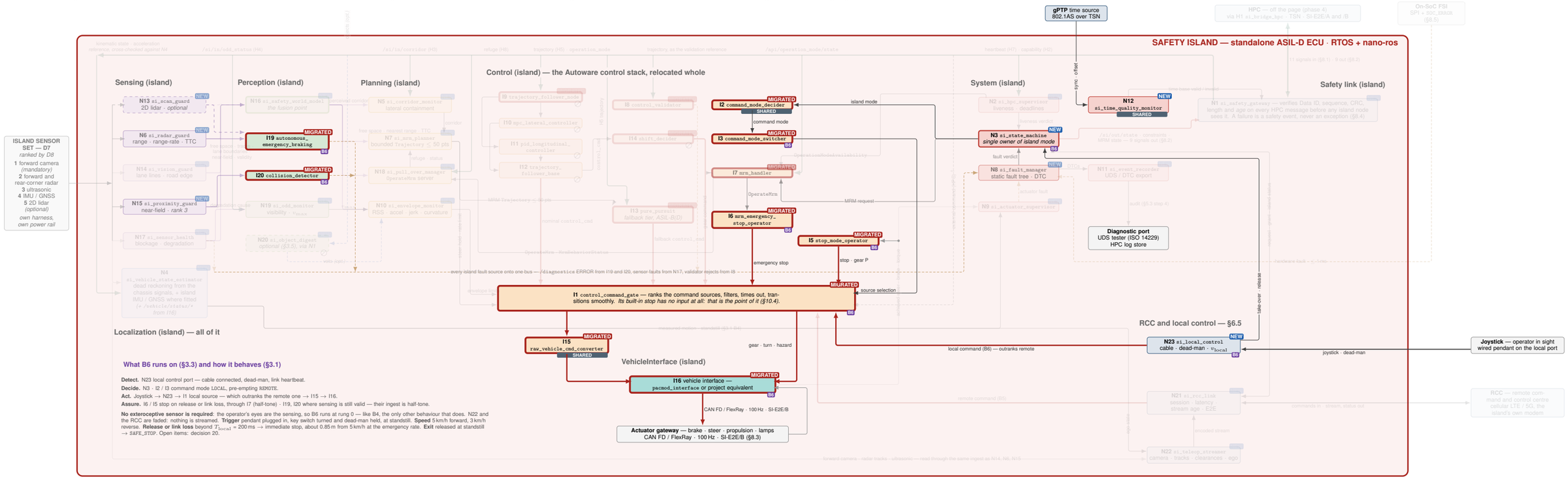
full colour — named for B5 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B5's path by §3.1, but not named in §3.3

faded — not used by B5. An edge is drawn at the weaker of its two ends. Only B5's own tag is drawn on each box.

Phase 5 · B6 — Manual joystick control without streamed sensing

An operator in sight of the vehicle drives it with a joystick on the island's local control port — by cable; a radio link is deferred to the next design — with no sensor stream; the island holds the speed to a creep and stops on release or on loss of the link.
The phase-5 drawing, unchanged: what B6 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Stage-1 detailed design: §6.6, entered from standstill.*



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Where each node came from

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1 B2 B3 B4 B5 B6

SHARED

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Messages

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- island sensor data
- control command into the gate
- fault · /diagnostics
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B6, manual joystick control without streamed sensing

full colour — named for B6 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B6's path by §3.1, but not named in §3.3

faded — not used by B6. An edge is drawn at the weaker of its two ends. Only B6's own tag is drawn on each box.